HIGH-LEVEL DESIGN & LOW-LEVEL DESIGN

FoodSaver

Surplus Food Donation and Redistribution System

Software Engineering – Design Documentation

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1. Introduction

This document describes the architecture, modules, data flow, database structure and processing logic of FoodSaver. The system connects surplus-food donors with verified recipient organizations; it does not perform physical delivery.

2. High-Level Design (HLD)

2.1 System Architecture

Donor / Recipient / Administrator
↓ Web or Mobile User Interface
↓ Application Layer
Authentication | Donation Management | Requests | Verification | Notifications
↓ Database Layer
Users | Donations | Requests | Verification | Status History

2.2 Main Modules

Module	Description
Authentication	Registration, login, logout and role-based access.
Recipient Verification	Submit organization details; admin approves/rejects.
Donation Management	Donors add, edit, view and cancel listings.
Search and Browse	Verified recipients search available donations.
Request Management	Recipients request food; donors accept/reject.
Notifications/History	Status alerts and completed/cancelled records.
Administration	Manage users, verification and donation records.

2.3 Data Flow

1. Donor registers and logs in. 2. Donor enters food details and submits a listing. 3. The application validates and stores it. 4. A verified recipient searches listings and submits a request. 5. The donor accepts or rejects the request. 6. The application updates status and notifies both parties. 7. The recipient arranges pickup and the record is marked completed when confirmed.

3. Low-Level Design (LLD)

3.1 Database Design

Table	Important fields
Users	user_id (PK), name, email, password_hash, phone, role, account_status
Recipient_Verification	verification_id (PK), user_id (FK), organization_name, proof_reference, status, reviewed_by
Donations	donation_id (PK), donor_id (FK), food_name, food_type, quantity, prepared_at, safe_until, pickup_location, status
Requests	request_id (PK), donation_id (FK), recipient_id (FK), requested_quantity, request_time, status
Notifications	notification_id (PK), user_id (FK), message, created_at, read_status

3.2 Processing Logic

Login: Accept email/password → validate fields → find user → verify password hash → check account status → open role-specific dashboard. Otherwise show an error.

Create donation: Confirm donor login → validate food name, positive quantity, safe-consumption time and pickup location → save listing with Available status → show confirmation.

Request donation: Confirm recipient is verified → check listing is available and not past its safe-consumption time → create Pending request → notify donor.

Accept/reject: Confirm donor owns the listing → check request is Pending → update status → notify recipient.

3.3 Validation and Security

Email must have a valid format; required fields cannot be empty; quantity must be greater than zero; safe-consumption time must be valid. Only verified recipients may request food. Passwords should be stored as secure hashes. Role-based authorization must restrict access to each user's permitted actions.

4. Non-Functional Design

The interface should be responsive and easy to use. Common operations should be quick. Database backups should be maintained. Modules should be separated to support testing and future maintenance.

5. Conclusion

The HLD defines the overall architecture and major modules. The LLD describes the database and processing logic. Together, they provide a blueprint for implementing FoodSaver.